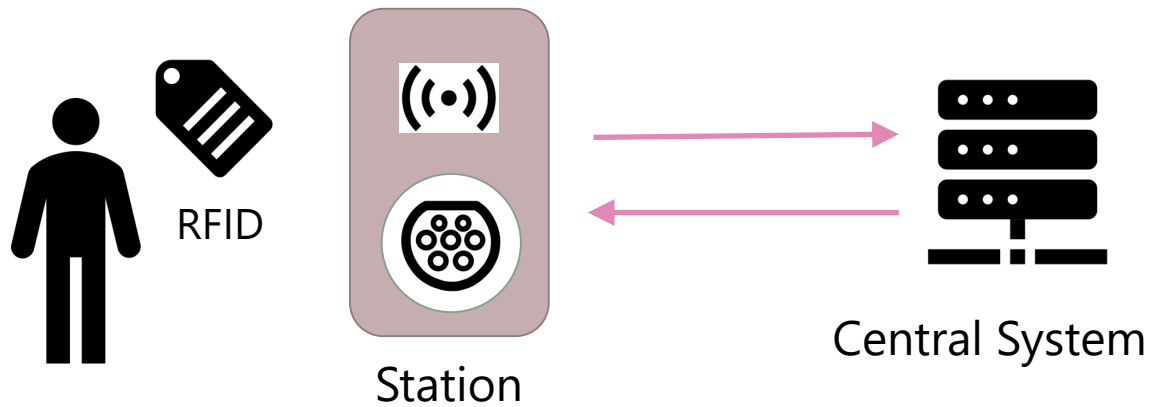


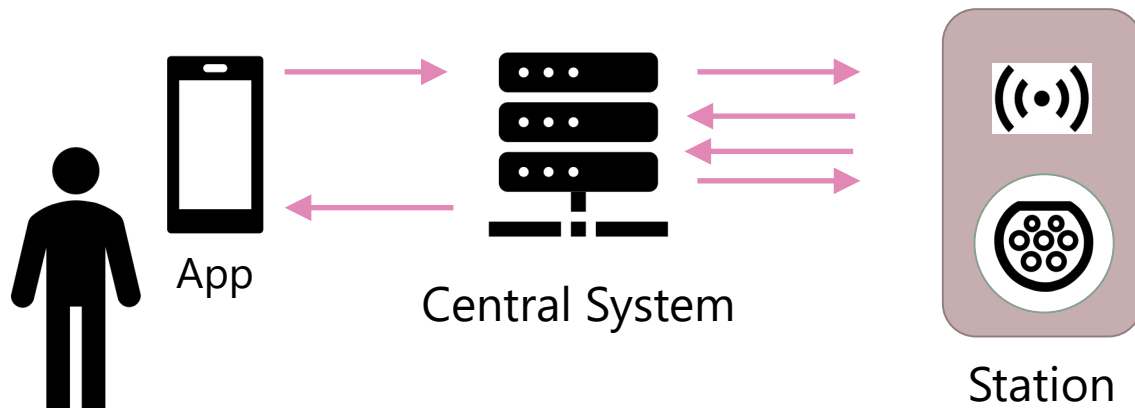
6. REMOTE START TRANSACTION



Mobile Apps and other UIs



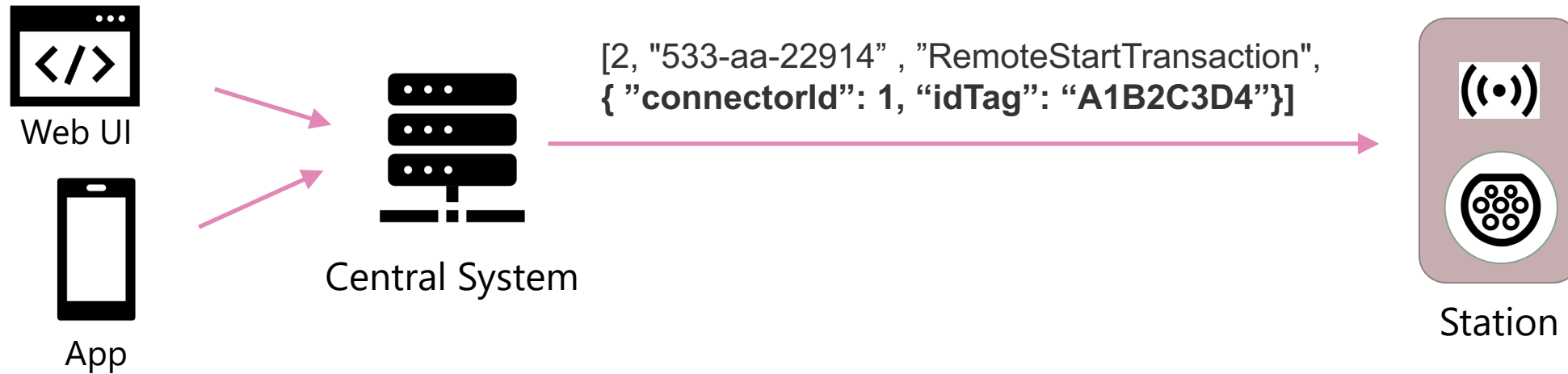
In the previous lesson, we discussed the basic charging flow when an RFID tag is used to start charging at a station: the station sends **Authorize** and **StartTransaction** messages to the central system, which then accepts or rejects them.



Another way to use stations is to start a charge through **a mobile app** or a remote interface such as **an admin system**. In these cases, the flow is slightly different: the central system initiates the charging with a **RemoteStartTransaction**.



Step 1: RemoteStartTransaction.req -message

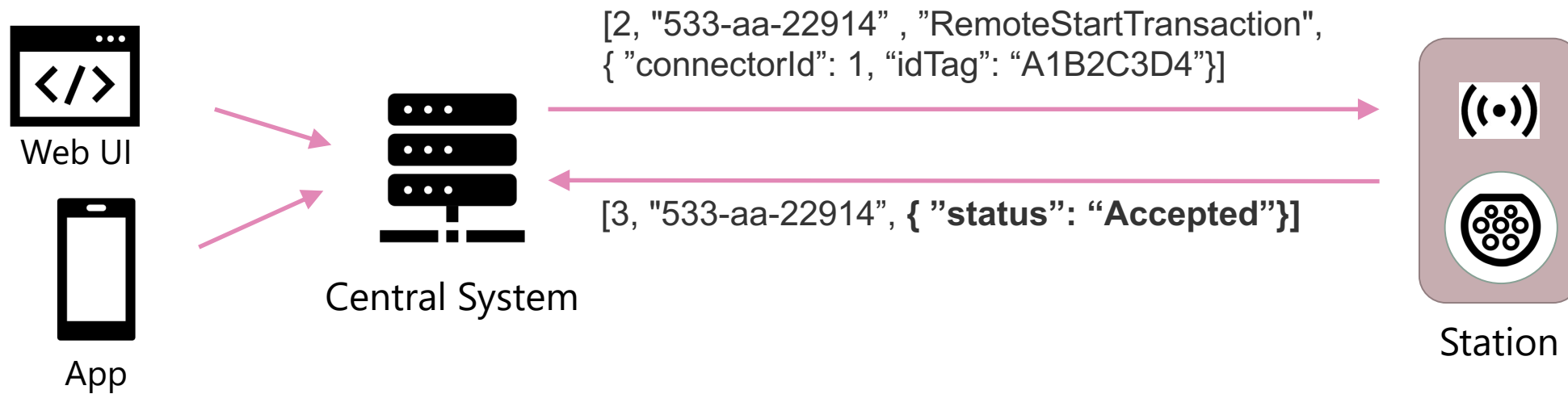


Mobile apps, admin systems, or other types of user interfaces for EV charging connect to the central system. The method for doing this is outside the scope of OCPP, but it could be done through a REST API, for example.

The central system authenticates and authorizes the request from an app. The method for doing this is also outside the scope of the OCPP protocol. Once the central system **has decided that it is ok** to start charging at the station, it sends a **RemoteStartTransaction.req** message to the station.



Step 2: RemoteStartTransaction.conf -message

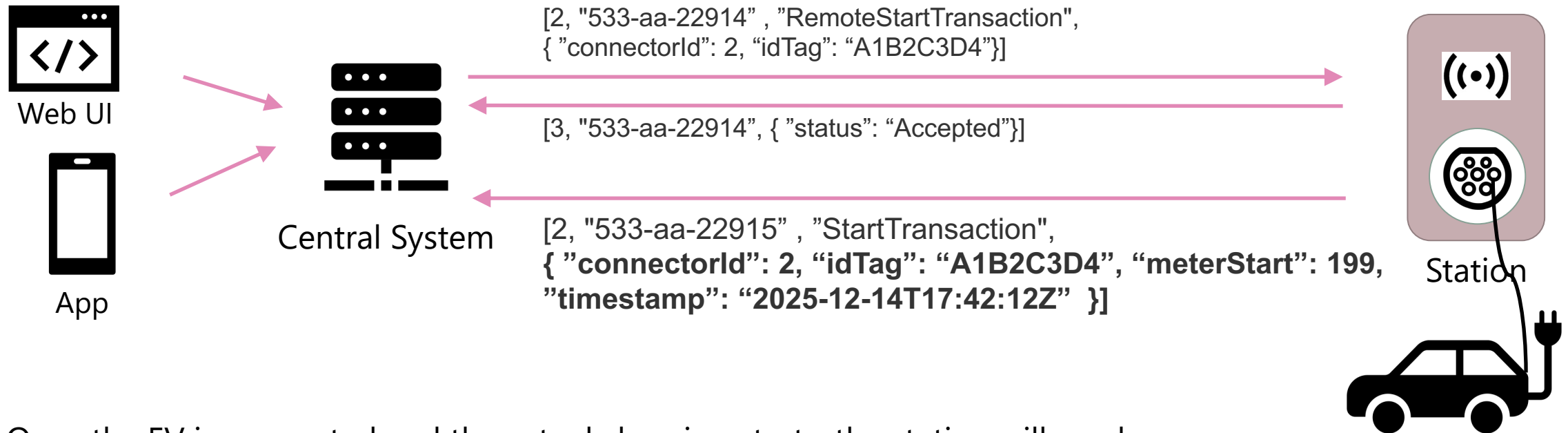


After receiving the request message, the station checks if a new transaction can be started and then responds with either "**Accepted**" or "**Rejected**" to the central system.

The station's "**Accepted**" response to the remote start request does not mean that charging has actually started; it only means that it is possible to start a new transaction. When the actual transaction begins, the station will send another message, **StartTransaction**, to the central system.



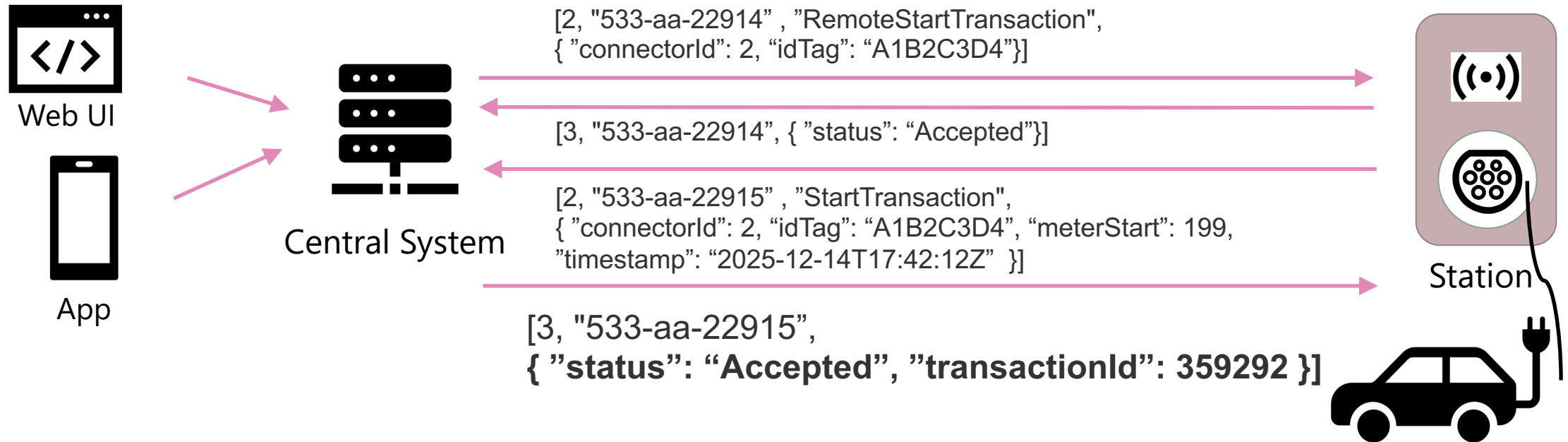
Step 3: StartTransaction.req-message



Once the EV is connected and the actual charging starts, the station will send a normal **StartTransaction.req** message to the central system. This follows the same structure and flow as we discussed in the previous lesson with RFID-tag charging.



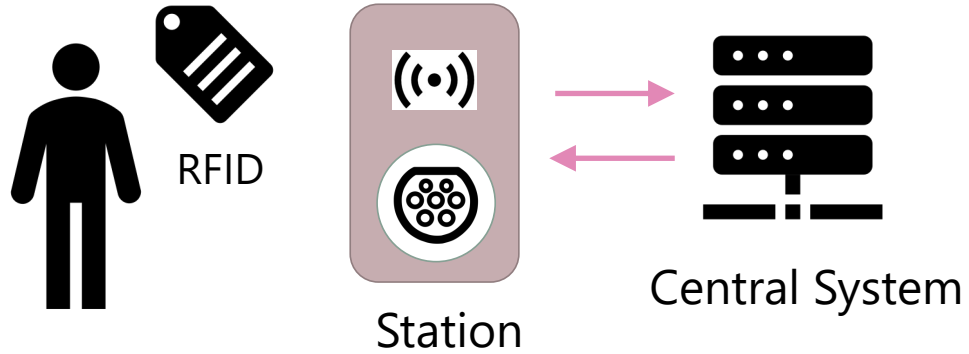
Step 4: StartTransaction.conf-message



As the final step of the flow, the central system then responds with a **StartTransaction.conf** message to the charging station, indicating which transaction ID the station should use for this transaction.

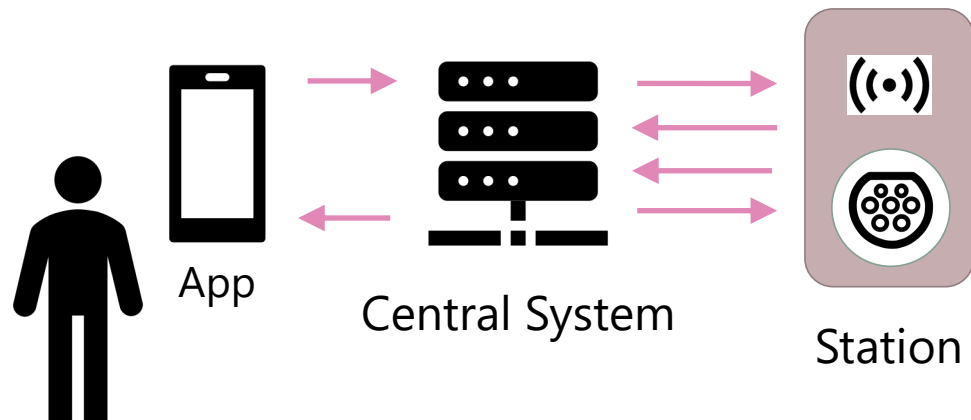


Summary: Remote vs. Local authorization



Local authorization flow:

1. An EV driver shows an RFID-tag to the station
2. The station sends **Authorize.req** to the Central System
3. If authorize is accepted, and an EV connected to the station, the station sends **StartTransaction.req** to the Central System



Remote authorization flow:

1. An EV driver starts charging from an app. The app calls an API in the Central System, which authorizes the request
2. The Central System sends **RemoteStartTransaction.req** to the station, which accepts or rejects this
3. If the station accepts and an EV is connected to the station, the station sends **StartTransaction.req** to the Central System

